Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Steady velocity

Meaning in my words How would you explain steady velocity to a friend?	**How can you tell?** Photographs are one second apart. What pattern would you look for?
Example or sketch Draw three positions for a cart moving 2 m right each second.	**Non-example and why** A bike circles a track at a steady speed. Is its velocity constant?

Constant acceleration

Meaning in my words How would you explain constant acceleration to a friend?	**How can you tell?** The velocities are 1, 3 and 5 m/s one second apart. What repeats?
Example or sketch If those gains continue, what velocity would you write for the next second?	**Non-example and why** Someone adds 2 m to position every second and calls that speeding up. What pattern did they draw?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Average velocity

Meaning in my words How would you explain average velocity to a friend?	**How can you tell?** A cart travels out and back. Which two points determine its average velocity?
Example or sketch Right is positive. A cart goes from +8 m to +2 m in 3 s. What is its average velocity?	**Non-example and why** “Average the two speeds, and you always get average velocity.” What is missing?

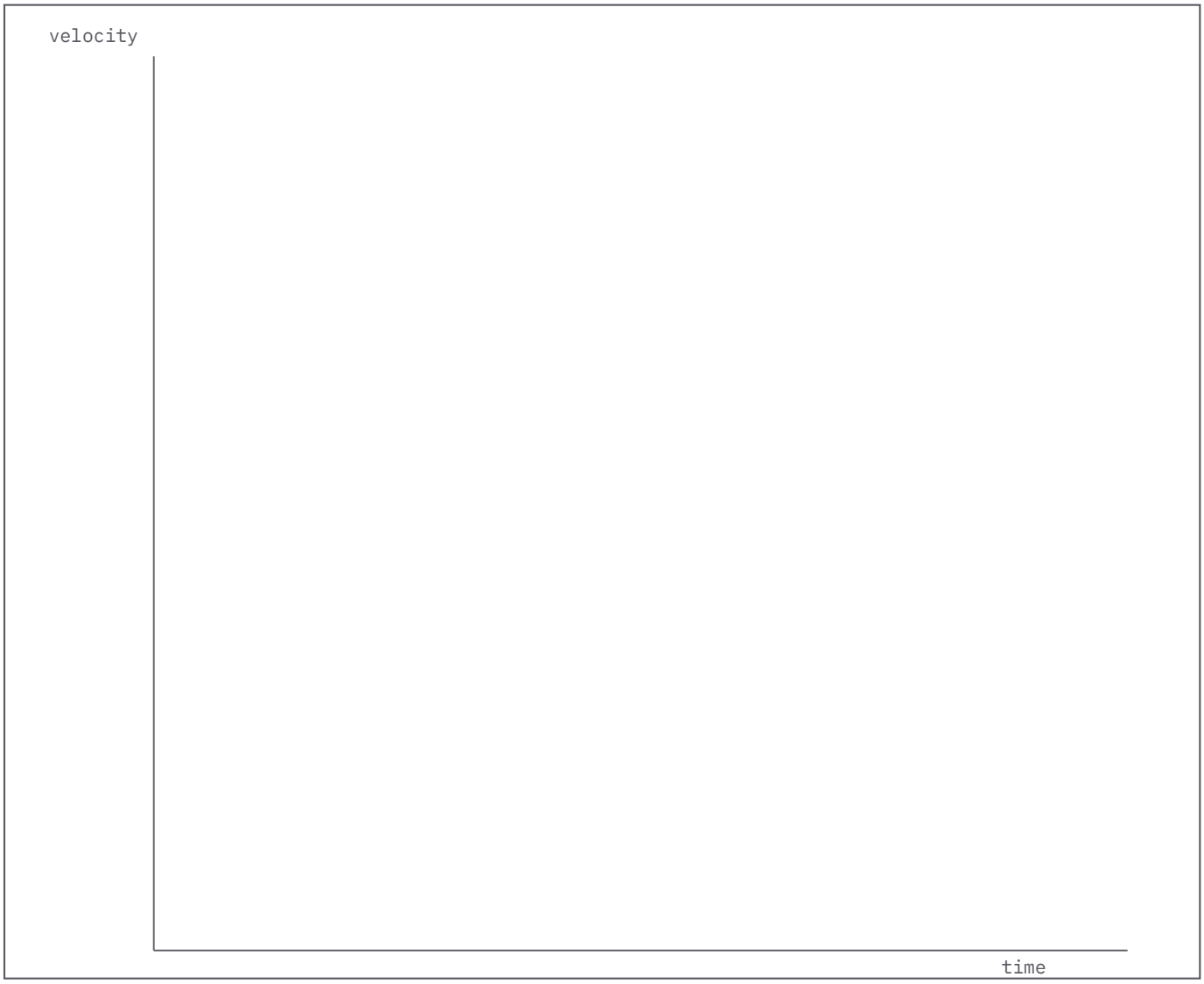
Model selection

Meaning in my words How would you explain model selection to a friend?	**How can you tell?** Which model fits velocity readings of 2, 4 and 6 m/s one second apart?
Example or sketch Which model would you use for a cart moving straight at 3 m/s every second?	**Non-example and why** A friend chooses a formula because it is the shortest. What should you ask first?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a straight velocity–time line. Label which feature gives acceleration and which gives displacement.



My marks mean

Catch the next shot

08

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity-time graph to predict displacement.

CHOOSE ONE JOB TOGETHER

CAMERA ENGINEER ☐

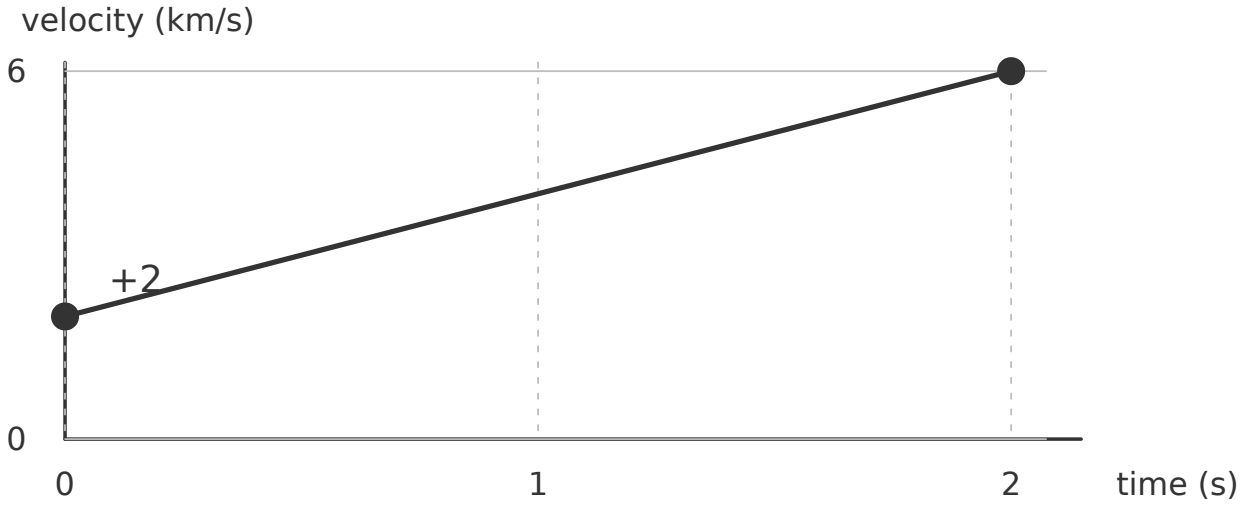
Shade the area under the velocity-time trace. Send the camera's displacement at the end of the shot.

REPLAY EDITOR ☐

Explain why using the final speed for the whole clip would miss the shot. Point to the graph as you speak.

Point to a clue. Keep your first marks; add the update using the color key.

XJ moves from **+2** to **+6 km/s** over **2 s**. The rehearsal assumes constant acceleration. Where should the next camera look relative to the start?



01 OUR FIRST CALL

Where should the camera look relative to the start? Show how the graph helps.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new report still let you guarantee that displacement? Explain your revised call.

Switch speakers. What does your shaded area measure?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks